# Stage 8 Science — Practice Test 2 — Mark Scheme

Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

## Paper 1 — 45 minutes — 50 marks

**1** Hassan feels tired and gets out of breath easily. At the hospital, a nurse takes a sample of his blood for a blood test. The table shows two of his results. **[1]**

| Component of blood | Hassan's result | Normal range for an adult man |
|---|---|---|
| red blood cells (millions per $mm^3$) | 3.6 | 4.5 to 5.9 |
| white blood cells (per $mm^3$) | 15 000 | 4 000 to 11 000 |

(a) Which component of Hassan's blood is **lower** than the normal range?

**Answer:** red blood cells or red blood cell or red cells or red or RBC or RBCs or erythrocytes
**Notes / working:** Red blood cells: 3.6 million per $mm^3$ is below the normal range of 4.5 to 5.9 million. (White blood cells are **higher** than normal, not lower.)

(b) Use your answer to (a) to explain why Hassan feels tired and gets out of breath. **[2]**

**Answer:** 1 mark: fewer red blood cells, so less oxygen is carried/transported (to his muscles/cells). 1 mark: so less (aerobic) respiration happens in his cells / less energy is released (so he is tired; he breathes faster to try to get more oxygen). Do not accept 'he has less blood' or 'red blood cells make energy'.
**Notes / working:** Model answer: "Red blood cells **carry oxygen**. Hassan has fewer of them, so **less oxygen reaches his cells**. His cells can carry out **less aerobic respiration**, so **less energy is released** and he feels tired. He breathes harder to try to get more oxygen." One mark for the oxygen idea, one for the link to respiration/energy.

(c) Hassan's white blood cell count is **higher** than normal. What does this most likely show? **[1]**

- ☐ A) He is not eating enough iron.
- ☐ B) His plasma is carrying too much carbon dioxide.
- ☐ C) His body is fighting an infection caused by pathogens.
- ☐ D) His lungs are not taking in enough air.

**Answer:** C) His body is fighting an infection caused by pathogens.
**Notes / working:** White blood cells **protect the body against pathogens** (microorganisms that cause disease). When the body is fighting an infection, it makes more white blood cells, so a high count is a sign of infection.

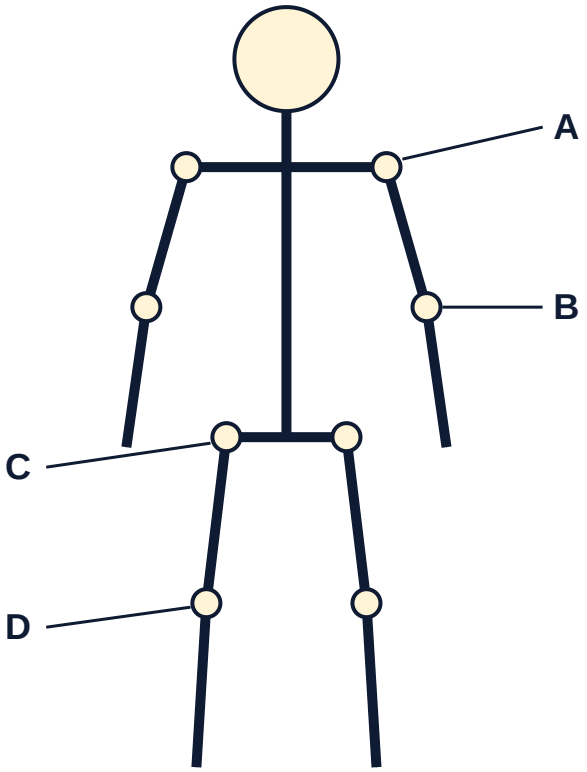
(d) The liquid part of the blood is called plasma. Which list shows things that plasma transports? **[1]**

- ☐ A) blood cells, nutrients and carbon dioxide
- ☐ B) air, bones and pathogens only
- ☐ C) oxygen joined to haemoglobin only
- ☐ D) white blood cells only

**Answer:** A) blood cells, nutrients and carbon dioxide
**Notes / working:** Plasma is the pale yellow liquid part of blood. It **transports blood cells, nutrients (such as glucose) and carbon dioxide**. Oxygen is carried by haemoglobin inside red blood cells, not mainly by plasma.

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**2** After a fall, Hassan has an X-ray. The diagram is a simple model of the human skeleton. A, B, C and D are joints. **[1]**




(a) Which **two** letters show **hinge** joints?

- ☐ A) A and C
- ☐ B) B and D
- ☐ C) A and B
- ☐ D) C and D

**Answer:** B) B and D
**Notes / working:** B is the **elbow** and D is the **knee**. They are hinge joints: they let the bones move backwards and forwards in one direction only, like a door on its hinges. A (shoulder) and C (hip) are **ball-and-socket** joints, which can move in almost every direction.

(b) Hassan's physiotherapist asks him to **straighten** his arm at joint B. Name the muscle that **contracts** to straighten the arm. **[1]**

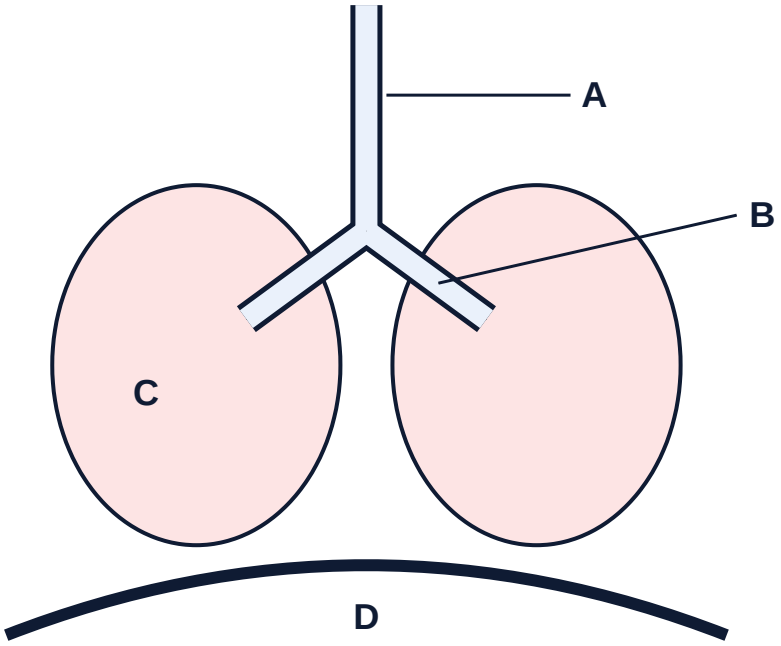
> **Answer:** triceps or tricep or the triceps or triceps muscle or tricepts
> **Notes / working:** The **triceps** (at the back of the upper arm) contracts and pulls the lower arm straight. At the same time the biceps relaxes.

(c) Explain why the muscles that move the arm at joint B must work as an **antagonistic pair**. **[2]**

> **Answer:** 1 mark: muscles can only pull (contract); they cannot push. 1 mark: so one muscle bends the arm (biceps) and the other straightens it (triceps) / when one contracts, the other relaxes. Do not accept 'muscles push the bones'.
> **Notes / working:** Model answer: "A muscle can only **pull** when it contracts; it cannot push. So two muscles are needed: the **biceps contracts to bend** the arm and the **triceps contracts to straighten** it. When one contracts, the other relaxes."

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**3** In the hospital, a doctor uses this diagram to explain the human respiratory system to Hassan. **[1]**




(a) Which letter shows the **diaphragm**?

> **Answer:** D
> **Notes / working: D** is the diaphragm: the dome-shaped sheet of muscle under the lungs. (A is the trachea, B is a bronchus, C is a lung.)

(b) Hassan breathes **in**. Which statement describes what happens? **[1]**

- ☐ A) The diaphragm relaxes and moves up, and the volume of the chest decreases.
- ☐ B) The intercostal muscles relax, so the ribs move up and out.
- ☐ C) The diaphragm contracts and moves up, so air is pushed out.
- ☐ D) The diaphragm contracts and flattens, and the volume of the chest increases.

**Answer:** D) The diaphragm contracts and flattens, and the volume of the chest increases.
**Notes / working:** Breathing in: the **diaphragm contracts and flattens** and the **intercostal muscles contract**, pulling the ribs up and out. The chest volume **increases**, the pressure inside falls, and air is pushed into the lungs by the higher air pressure outside.

(c) Oxygen moves from the air in the alveoli into the blood. Describe how. **[2]**

**Answer:** Any two of: (by) diffusion; oxygen moves from a high concentration (in the alveoli) to a low concentration (in the blood); through the thin (one-cell-thick) walls of the alveoli and capillaries. Do not accept 'oxygen is pumped into the blood' or 'osmosis'.
**Notes / working:** Model answer: "Oxygen moves by **diffusion**. The concentration of oxygen is **higher in the air in the alveoli than in the blood**, so oxygen moves from high to low concentration, through the **very thin walls** of the alveoli and the capillaries." Carbon dioxide diffuses the other way, from the blood into the alveoli.

(d) Hassan says, "Breathing and respiration are the same thing." Which statement shows that he is **wrong**? **[1]**

- ☐ A) Breathing happens only in animals; respiration happens only in plants.
- ☐ B) Breathing moves air into and out of the lungs; respiration is a reaction in cells that releases energy.
- ☐ C) Breathing releases energy from glucose; respiration moves air into the lungs.
- ☐ D) Breathing uses carbon dioxide; respiration makes oxygen.

**Answer:** B) Breathing moves air into and out of the lungs; respiration is a reaction in cells that releases energy.
**Notes / working: Breathing** is the movement of air in and out of the lungs (a physical process). **Respiration** is a chemical reaction in every living cell that releases energy from glucose. Plants and animals both respire.

**4** A food factory makes a breakfast cereal. This is part of the nutrition label. **[1]**

| Nutrient | Amount in 100 g of cereal |
|---|---|
| energy | 1 600 kJ |
| protein | 9 g |
| carbohydrate | 76 g |
| fat | 3 g |
| calcium | 0.5 g |
| iron | 0.012 g |
| vitamin D | added |

(a) Which nutrient in the table gives this cereal **most** of its energy?

**Answer:** carbohydrate or carbohydrates or carbs or carbohydrate (starch) or starch or carbohydrates and sugar or sugar and starch
**Notes / working: Carbohydrate.** There are 76 g of carbohydrate in 100 g, much more than protein (9 g) or fat (3 g), so most of the energy comes from carbohydrate. (Gram for gram, fat stores more energy, but there is very little fat here.)

(b) Ali eats a 40 g bowl of the cereal. Calculate the energy in his bowl of cereal. Show your working. **[2]**

**Answer:** 640 kJ
**Notes / working:** 1 600 kJ is in 100 g, so 1 g gives 1 600 ÷ 100 = 16 kJ. 40 g gives 16 × 40 = **640 kJ**. Mark scheme: M1 for 1 600 ÷ 100 × 40 (or 1 600 × 0.4) seen; A1 for 640 (kJ).

(c) Vitamin D is added to the cereal. What is the function of vitamin D? **[1]**

- ☐ A) It is needed to make haemoglobin for red blood cells.
- ☐ B) It is the main store of energy in the body.
- ☐ C) It helps the body take in calcium for strong bones and teeth.
- ☐ D) It is used to build new muscle tissue for growth.

**Answer:** C) It helps the body take in calcium for strong bones and teeth.
**Notes / working: Vitamin D helps the body absorb calcium**, which is needed for strong bones and teeth. Iron is needed to make haemoglobin; fat and carbohydrate are energy stores; protein is for growth and repair.

**5** In hospital, patients with lung disease are sometimes given extra oxygen through a mask. Their cells use the oxygen for aerobic respiration. **[1]**

(a) Complete the word equation for aerobic respiration.
glucose + oxygen → _____ + water

**Answer:** carbon dioxide or CO2 or $CO_2$ or carbondioxide
**Notes / working:** glucose + oxygen → **carbon dioxide** + water. The carbon dioxide is carried by the plasma to the lungs and breathed out.

(b) In which part of a cell does aerobic respiration take place? **[1]**

- ☐ A) the nucleus
- ☐ B) the cell membrane
- ☐ C) the cell wall
- ☐ D) the mitochondria

**Answer:** D) the mitochondria
**Notes / working:** Aerobic respiration takes place in the **mitochondria**, in both plant and animal cells. The nucleus controls the cell; the membrane controls what goes in and out; animal cells have no cell wall.

(c) Aerobic respiration gives a **controlled** release of energy. Explain what "controlled" means here. **[1]**

**Answer:** 1 mark: the energy is released slowly / gradually / in small steps (controlled by enzymes), not all at once as in burning. Do not accept 'energy is made' or 'energy is controlled by the brain'.
**Notes / working:** Model answer: "The energy is released **slowly, in many small steps**, not all at once like when glucose burns in a flame." This means cells can use the energy without overheating.

**6** Lake Nasser is a large lake behind the Aswan High Dam. Its ecosystem includes open water, the muddy lake bed, reed beds near the shore and small rocky islands. **[1]**

(a) Which of these is a **habitat** within the Lake Nasser ecosystem?

- ☐ A) the reed beds near the shore
- ☐ B) all the living things in the lake
- ☐ C) the Nile crocodile
- ☐ D) the amount of sunlight that reaches the water

**Answer:** A) the reed beds near the shore
**Notes / working:** A **habitat** is the place where an organism lives, for example the reed beds, the muddy lake bed or the open water. 'All the living things' is a community, a crocodile is an organism, and sunlight is a physical factor.

(b) Water hyacinth is a plant that came from South America. It now grows quickly on parts of the River Nile and covers the surface of the water. Describe **two** ways it can affect other organisms in the ecosystem. **[2]**

> **Answer:** Any two of: it blocks light, so plants/algae under the water cannot photosynthesise (and die); less oxygen in the water (as plants die and rot), so fish/animals die; it competes with native plants for space/nutrients/light; few/no animals eat it (no predators), so it keeps spreading; food chains/webs are disrupted (animals that eat native plants lose food). 1 mark each. Do not accept 'it is bad for the environment' with no detail.
> **Notes / working:** Model answer: "(1) The thick mat of leaves **blocks sunlight**, so plants under the water cannot photosynthesise and die. (2) When these plants die and rot, **oxygen in the water falls**, so fish may die." Other good points: it competes with native plants; nothing eats it, so it spreads fast.

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**7** Pesticide from farms near Lake Nasser washes into the water. Scientists measured the concentration of the pesticide in organisms in one food chain (example data). **[1]**

| Organism | Concentration of pesticide in parts per million (ppm) |
|---|---|
| water plants | 0.04 |
| small fish | 0.6 |
| Nile perch | 3 |
| fish eagle | 30 |

(a) How many times greater is the concentration of pesticide in the fish eagle than in the small fish?

> **Answer:** 50 times
> **Notes / working:** 30 ÷ 0.6 = **50 times** greater.

(b) Explain why the fish eagle has the highest concentration of pesticide. **[2]**

> **Answer:** 1 mark: the pesticide is not broken down / not excreted, so it stays (is stored) in the body / builds up in the organism. 1 mark: each animal eats many organisms from the level below, so the pesticide builds up (accumulates) at each step / the eagle is at the top of the food chain. Accept 'bioaccumulation' for the first mark only if explained.
> **Notes / working:** Model answer: "The pesticide **is not broken down or removed** from the body, so it stays in the tissues. Each small fish eats many plants, each perch eats many small fish and each eagle eats many perch, so the pesticide **builds up at every step** of the food chain. The eagle is at the top, so it has the most." This is bioaccumulation.

**8** Solar panels are made from silicon. An atom of silicon has 14 protons, 14 neutrons and 14 electrons. **[1]**

(a) Complete the sentence.
Protons have a _____ charge, but neutrons have no charge.

**Answer:** positive or + or +1 or positive charge or plus
**Notes / working:** Protons are **positive**, electrons are negative and neutrons have no charge (they are neutral).

(b) Which sentence describes the **Rutherford model** of the atom? **[1]**

- ☐ A) A ball of positive charge with negative electrons spread all through it.
- ☐ B) A solid ball with no charge and no smaller parts inside it.
- ☐ C) A tiny positive nucleus in the centre, with negative electrons around it and mostly empty space.
- ☐ D) A large negative nucleus with positive electrons moving around it.

**Answer:** C) A tiny positive nucleus in the centre, with negative electrons around it and mostly empty space.
**Notes / working:** Rutherford's model: almost all the mass is in a **tiny, positive nucleus** at the centre, with **electrons** around it, and **most of the atom is empty space**. Option A is the older 'plum pudding' model.

(c) A teacher says: "If an atom were as big as a football stadium, its nucleus would be about the size of a pea on the centre spot." What does this analogy show? **[1]**

- ☐ A) Atoms are as large as football stadiums.
- ☐ B) The nucleus is very small compared with the whole atom, so most of the atom is empty space.
- ☐ C) The nucleus is round and green like a pea.
- ☐ D) Electrons are much bigger than the nucleus.

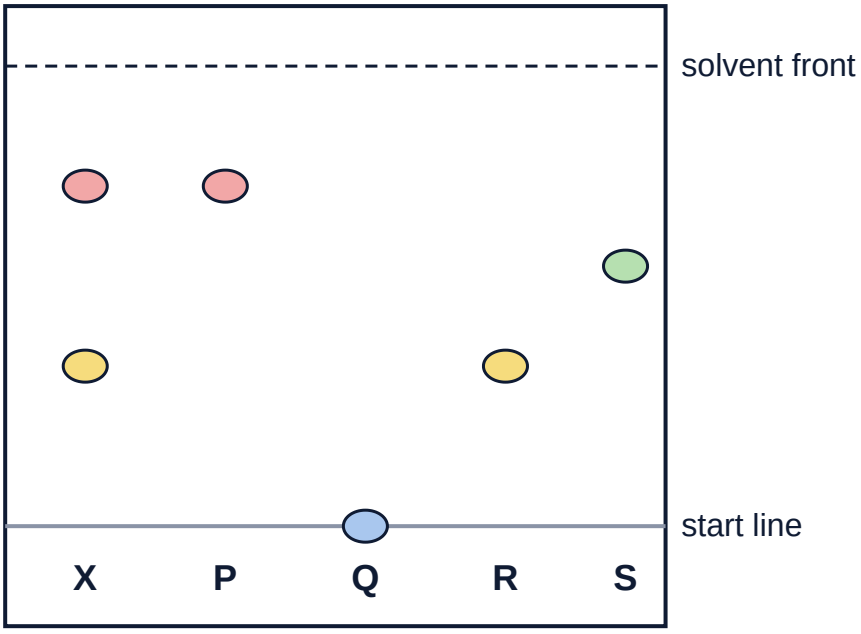
**Answer:** B) The nucleus is very small compared with the whole atom, so most of the atom is empty space.
**Notes / working:** An **analogy** compares something we cannot see with something familiar. The pea and the stadium show the **sizes compared**: the nucleus is tiny and the rest of the atom is mostly empty space. The analogy does not mean an atom is really that big or that colour.

(d) Explain what holds the electrons in the atom. **[1]**

> **Answer:** 1 mark: (electrostatic) attraction between the negative electrons and the positive nucleus/protons. Need both the idea of attraction and opposite charges. Do not accept 'gravity' or 'magnetism'.
> **Notes / working:** Model answer: "The **negative electrons are attracted to the positive protons** in the nucleus. This electrostatic attraction between opposite charges holds the atom together."

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**9** In the food factory's laboratory, a scientist tests the colouring in a sweet, X. She compares it with four food colours, P, Q, R and S, that are allowed in food. The diagram shows her chromatogram. **[1]**




(a) Which **two** food colours are in sweet X?

> **Answer:** P and R or R and P or P, R or R, P or P & R or R & P or PR or RP
> **Notes / working:** X separates into two spots. One is at the same height as P and the other is at the same height as R, so X contains **P and R**.

(b) Food colour Q did **not** move up the paper. Suggest why. **[1]**

> **Answer:** 1 mark: Q is insoluble in the solvent / does not dissolve in the solvent (used). Do not accept 'Q is too heavy' or 'Q is pure'.
> **Notes / working:** A substance only moves up the paper if it **dissolves in the solvent**. Q stayed on the start line, so it is **insoluble in this solvent**.

(c) Why must the start line be **above** the level of the solvent in the beaker? **[1]**

- ☐ A) So that the solvent moves up the paper faster.
- ☐ B) So that the paper does not get wet.
- ☐ C) So that the pencil line does not move.
- ☐ D) So that the colours do not dissolve straight into the solvent in the beaker.

**Answer:** D) So that the colours do not dissolve straight into the solvent in the beaker.
**Notes / working:** If the spots were below the solvent, the colours would **wash off into the solvent in the beaker** instead of moving up the paper and separating.

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**10** (continued) **[2]**

The factory also buys salt. A 50 g sample of the salt contains 48 g of sodium chloride. The rest is other substances.

(d) Calculate the percentage purity of the salt. Show your working.

**Answer:** 96 %
**Notes / working:** Percentage purity = mass of the substance you want ÷ total mass × 100 = 48 ÷ 50 × 100 = **96%**. Mark scheme: M1 for 48 ÷ 50 (× 100) seen; A1 for 96(%).

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**11** Omar is a diver at a swimming pool. His weight is 720 N. **[2]**

(a) Omar stands on the side of the pool. The total area of his feet touching the floor is 0.04 $m^2$. Calculate the pressure on the floor. Show your working.

**Answer:** 18000 $N/m^2$
**Notes / working:** pressure = force ÷ area = 720 ÷ 0.04 = **18 000 N/m²** (18 000 Pa). Mark scheme: M1 for 720 ÷ 0.04 seen; A1 for 18 000. Common mistake: 720 × 0.04 = 28.8 (multiplying instead of dividing).

(b) Omar puts on his diving flippers and stands on the floor again. Which statement is correct? **[1]**

- ☐ A) The pressure increases because his weight is bigger.
- ☐ B) The pressure stays the same because his weight has not changed.
- ☐ C) The pressure decreases because the same force acts on a bigger area.
- ☐ D) The pressure decreases because his weight decreases.

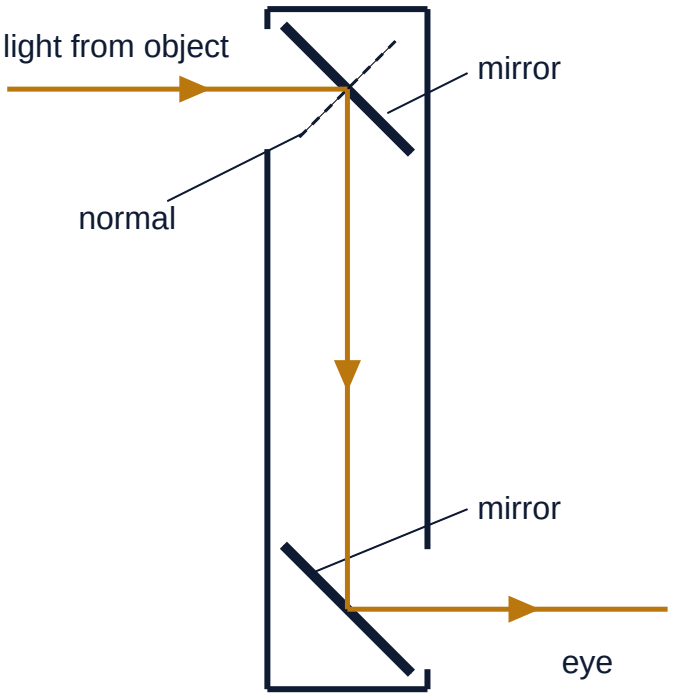
**Answer:** C) The pressure decreases because the same force acts on a bigger area.
**Notes / working:** His weight (the force) is the same, but flippers have a **bigger area**. pressure = force ÷ area, so a bigger area gives a **smaller pressure**.

(c) Omar's tank of air is left in hot sunshine before a dive. The pressure of the air inside the tank increases. Use ideas about particles to explain why. **[2]**

> **Answer:** 1 mark: the (air/gas) particles move faster / have more (kinetic) energy when they are hotter. 1 mark: they hit (collide with) the walls of the tank more often and/or with more force (harder). Do not accept 'the particles get bigger/expand' or 'there are more particles'.
> **Notes / working:** Model answer: "When the air is heated, its particles **move faster**. They **collide with the inside walls of the tank more often and with more force**, so the pressure increases." The number of particles and the volume of the tank stay the same.

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**12** Laila makes a periscope so that she can see over the wall of the swimming pool. It has two plane mirrors. Each mirror is at 45° to the sides of the tube. The diagram shows a ray of light passing through it. **[1]**




(a) What is the angle of incidence at the top mirror?

> **Answer:** 45 °
> **Notes / working:** The ray is horizontal and the mirror is at 45° to the sides of the tube, so the ray meets the mirror surface at 45°. The normal is at 90° to the mirror, so the angle of incidence = 90° − 45° = **45°**. The angle of reflection is also 45°, so the ray turns through 90° and travels down the tube.

**13** (continued) **[1]**

In a different mirror, a ray of light hits the mirror surface. The angle between the ray and the **mirror surface** is 35°.

(b) What is the angle of reflection?

**Answer:** 55 °
**Notes / working:** Angles are measured from the **normal**, not from the mirror. Angle of incidence = 90° − 35° = 55°. By the law of reflection, angle of reflection = **55°**. Common mistake: answering 35°.

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**14** Laila makes a periscope so that she can see over the wall of the swimming pool. It has two plane mirrors. Each mirror is at 45° to the sides of the tube. The diagram shows a ray of light passing through it. **[1]**

(c) State the law of reflection.

**Answer:** 1 mark: the angle of incidence equals the angle of reflection (accept i = r). Do not accept 'light bounces back' or 'the angles are the same' with no names.
**Notes / working: The angle of incidence = the angle of reflection**, both measured from the normal.

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**15** The food factory uses an electromagnet above a conveyor belt to pull small pieces of iron out of the cereal. Engineers make a model electromagnet and investigate how the **current** affects the **mass of iron filings** it can lift. They do each test three times. **[1]**

| Current in A | Mass lifted (1st) in g | Mass lifted (2nd) in g | Mass lifted (3rd) in g |
|---|---|---|---|
| 0.5 | 4.1 | 3.9 | 4.0 |
| 1.0 | 8.2 | 7.9 | 8.0 |
| 1.5 | 12.1 | 11.8 | 12.1 |
| 2.0 | 16.0 | 11.2 | 16.3 |
| 2.5 | 19.8 | 20.1 | 20.0 |

(a) Which equipment do the engineers need to make the electromagnet?

- ☐ A) an iron nail, a coil of insulated copper wire and a power supply
- ☐ B) a plastic rod, bare copper wire and a thermometer
- ☐ C) a bar magnet, a glass rod and a power supply
- ☐ D) an iron nail, a rubber band and a compass

**Answer:** A) an iron nail, a coil of insulated copper wire and a power supply
**Notes / working:** An electromagnet is a **coil of insulated wire** wrapped around an **iron core**, connected to a **power supply** so that a current flows. The iron core makes it much stronger. The wire must be insulated so the current goes around the coil, not through the nail.

(b) Name the **independent** variable in this investigation. **[1]**

> **Answer:** current or the current or electric current or current in A or size of the current or the size of the current or amps or amperes
> **Notes / working:** The **independent variable** is the one the engineers choose and change: the **current**. The dependent variable (what they measure) is the mass of iron filings lifted.

(c) At which current is there an anomalous result? **[1]**

> **Answer:** 2 or 2.0 A
> **Notes / working:** At **2.0 A**, the second result (11.2 g) is far below the other two results (16.0 g and 16.3 g) and does not fit the pattern. It is anomalous.

(d) Calculate the mean mass lifted at 2.0 A. Do **not** include the anomalous result. **[1]**

> **Answer:** 16.15 g
> **Notes / working:** Leave out 11.2 g. Mean = (16.0 + 16.3) ÷ 2 = 32.3 ÷ 2 = **16.15 g** (accept 16.1 g or 16.2 g). Common mistake: including 11.2 g, which gives 14.5 g.

(e) Describe the relationship between the current and the mass of iron filings lifted. **[1]**

> **Answer:** 1 mark: as the current increases, the mass lifted increases. Accept 'the mass is (directly) proportional to the current' or 'doubling the current doubles the mass'. Do not accept 'the mass changes' with no direction.
> **Notes / working:** Model answer: "**As the current increases, the mass lifted increases.** When the current doubles (0.5 A → 1.0 A, or 1.0 A → 2.0 A) the mass doubles (about 4 g → 8 g → 16 g), so the mass is proportional to the current."

(f) Explain why the factory uses an electromagnet and not a permanent magnet to remove the iron. **[1]**

> **Answer:** 1 mark: an electromagnet can be switched off, so the iron pieces drop off / are released (into a bin) / it can be cleaned easily. Accept 'its strength can be changed (by changing the current)'. Do not accept 'it is stronger' alone.
> **Notes / working:** An electromagnet is only magnetic when the current flows. The factory can **switch it off so that the pieces of iron fall off** into a waste bin. A permanent magnet would keep holding them.

## Paper 2 — 45 minutes — 50 marks

**1** A hospital first-aid kit has an instant cold pack and a hand warmer. When the cold pack is squeezed, a solid inside it dissolves in water. A nurse measures the temperature of the cold pack. **[1]**

| Time in minutes | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Temperature in °C | 22 | 12 | 7 | 6 |

(a) By how much does the temperature of the cold pack fall in 3 minutes?

**Answer:** 16 °C
**Notes / working:** 22 − 6 = **16 °C**.

(b) Which word describes the process in the cold pack? **[1]**

- ☐ A) endothermic
- ☐ B) exothermic
- ☐ C) combustion
- ☐ D) neutralisation

**Answer:** A) endothermic
**Notes / working:** The temperature **falls**, so the process takes in thermal energy from the surroundings. It is **endothermic**.

**2** (continued) **[1]**
The hand warmer contains iron powder. When the packet is opened, the iron reacts with oxygen in the air to make iron oxide, and the packet gets warm.

(c) Write the word equation for this reaction.

**Answer:** iron + oxygen → iron oxide or iron + oxygen -> iron oxide or iron + oxygen --> iron oxide or iron + oxygen = iron oxide or iron + oxygen => iron oxide or oxygen + iron → iron oxide or oxygen + iron -> iron oxide or oxygen + iron --> iron oxide or oxygen + iron = iron oxide or oxygen + iron => iron oxide
**Notes / working: iron + oxygen → iron oxide**. Reactants go on the left of the arrow and the product on the right. Do not write '=' in a real exam; use an arrow.

(d) How does the temperature change show that the reaction in the hand warmer is **exothermic**? **[1]**

**Answer:** 1 mark: the temperature (of the packet/surroundings) increases, because thermal energy (heat) is given out / transferred to the surroundings. Do not accept 'it uses heat' or 'it takes in energy'.
**Notes / working:** The temperature **rises**: the reaction **gives out thermal energy** to the surroundings (your hands). That is what exothermic means.

**3** A hospital runs a healthy-lifestyle clinic for parents. **[1]**

(a) Animals eat food to get energy and nutrients. Which **two** nutrients can the human body store as an energy store?

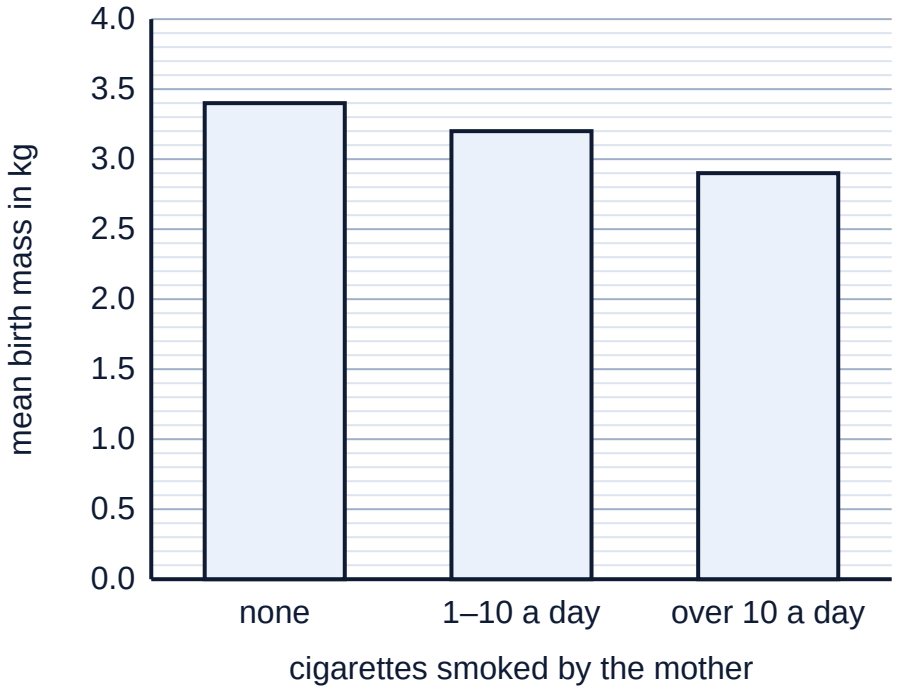
- ☐ A) vitamins and minerals
- ☐ B) water and protein
- ☐ C) calcium and iron
- ☐ D) carbohydrates and fats

**Answer:** D) carbohydrates and fats
**Notes / working:** Animals store energy as **carbohydrate** (a small store in the liver and muscles) and, mainly, as **fat** (under the skin and around organs). Vitamins, minerals and water do not store energy.

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**4** (continued) **[1]**

A hospital study recorded the mean birth mass of babies born to three groups of mothers. The bar chart shows the results (example data).




(b) What was the mean birth mass of babies whose mothers smoked **more than 10** cigarettes a day?

**Answer:** 2.9 kg
**Notes / working:** The top of the third bar is at **2.9 kg** (one small square below 3.0).

(c) Describe the pattern shown by the bar chart. **[1]**

> **Answer:** 1 mark: the more cigarettes the mother smoked, the lower the (mean) birth mass (of the baby). Accept 'babies of smokers are lighter / smaller' only if it compares groups. Do not accept 'smoking is bad'.
> **Notes / working:** Model answer: "**The more cigarettes the mother smoked, the lower the mean birth mass** of her baby." (3.4 kg → 3.2 kg → 2.9 kg.)

(d) Which substance in tobacco smoke makes people **addicted** to smoking? **[1]**

- ☐ A) tar
- ☐ B) carbon monoxide
- ☐ C) nicotine
- ☐ D) oxygen

> **Answer:** C) nicotine
> **Notes / working: Nicotine** is the addictive drug in tobacco. Tar causes cancer and coats the lungs; carbon monoxide reduces the oxygen the blood can carry.

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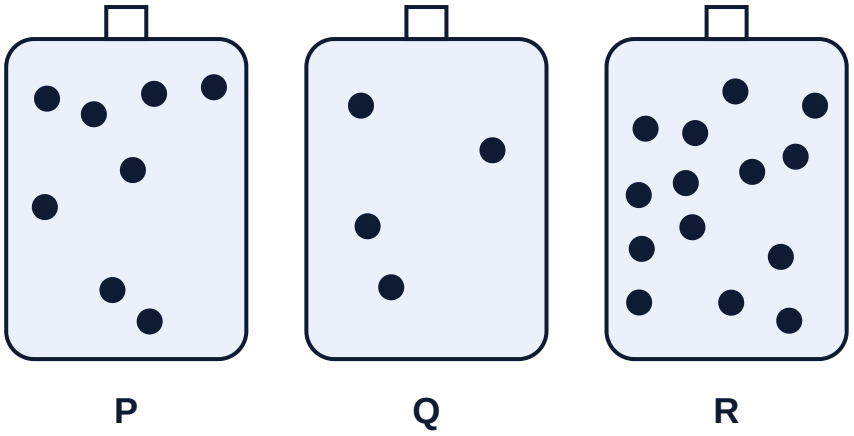
**5** (continued) **[1]**

A leaflet paid for by a tobacco company says: "Smoking a few cigarettes a day does no harm to your health."

(e) Explain why this source of information may be **biased**.

> **Answer:** 1 mark: the company makes money from selling cigarettes / wants people to keep buying cigarettes, so it may leave out or ignore evidence that smoking causes harm. Do not accept 'it is wrong' with no reason.
> **Notes / working:** A **biased** source is one that is one-sided because of who made it. The tobacco company **earns money when people smoke**, so it may only report information that helps sales. A hospital study or a health organisation would be a more reliable source.

**6** In hospital, some patients are given a salt solution through a drip. The diagrams show three drip bags, P, Q and R. Each bag holds the same volume of solution. Each dot represents a number of salt particles. **[1]**




(a) Which bag holds the **most concentrated** solution?

**Answer:** R
**Notes / working:** All the bags have the same volume. **R** has the most salt particles (14 dots, compared with 8 in P and 4 in Q), so it is the most concentrated.

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**7** (continued) **[1]**

A drip solution is made by dissolving 9 g of salt in water to make 1000 $cm^3$ of solution.

(b) What mass of salt is in 250 $cm^3$ of this solution?

**Answer:** 2.25 g
**Notes / working:** 250 $cm^3$ is one quarter of 1000 $cm^3$, so it contains one quarter of the salt: 9 ÷ 4 = **2.25 g**.

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**8** In hospital, some patients are given a salt solution through a drip. The diagrams show three drip bags, P, Q and R. Each bag holds the same volume of solution. Each dot represents a number of salt particles. **[1]**

(c) A nurse adds pure water to bag P. Use ideas about particles to explain why the solution becomes **less** concentrated.

**Answer:** 1 mark: the number of salt (solute) particles stays the same but they are in a bigger volume (of water), so there are fewer salt particles in each $cm^3$. Do not accept 'the salt particles get smaller' or 'the salt is destroyed'.
**Notes / working:** Model answer: "The **same number of salt particles** is now spread through a **bigger volume** of water, so there are **fewer salt particles in each $cm^3$**."

**9** A food factory uses two salts: sodium chloride (table salt) and potassium nitrate (a preservative used in some meat products). The table shows the mass of each salt that dissolves in 100 g of water at different temperatures. **[1]**

| Temperature in °C | 0 | 20 | 40 | 60 | 80 |
|---|---|---|---|---|---|
| sodium chloride in g | 36 | 36 | 37 | 37 | 38 |
| potassium nitrate in g | 13 | 32 | 64 | 110 | 169 |

(a) What mass of potassium nitrate dissolves in 100 g of water at 40 °C?

**Answer:** 64 g
**Notes / working:** Read across the potassium nitrate row to the 40 °C column: **64 g**.

(b) A worker dissolves as much potassium nitrate as possible in 100 g of water at 60 °C. She then cools the solution to 20 °C. Calculate the mass of potassium nitrate crystals that form. Show your working. **[2]**

**Answer:** 78 g
**Notes / working:** At 60 °C, 110 g dissolves. At 20 °C, only 32 g can stay dissolved. Crystals = 110 − 32 = **78 g**. Mark scheme: M1 for 110 and 32 read correctly / 110 − 32 seen; A1 for 78 (g).

(c) Which statement is correct? **[1]**

- ☐ A) The solubility of both salts increases a lot as the temperature rises.
- ☐ B) The solubility of sodium chloride changes very little as the temperature rises.
- ☐ C) Sodium chloride is more soluble than potassium nitrate at every temperature.
- ☐ D) The solubility of potassium nitrate decreases as the temperature rises.

**Answer:** B) The solubility of sodium chloride changes very little as the temperature rises.
**Notes / working:** Sodium chloride only goes from 36 g to 38 g between 0 °C and 80 °C, so its solubility **changes very little**. Potassium nitrate's solubility increases a lot. Sodium chloride is more soluble only at low temperatures (0 °C and 20 °C in the table); at 40 °C and above, potassium nitrate is far more soluble.

**10** (continued) **[1]**
When the factory's supplier makes potassium nitrate by a chemical reaction, the product is often impure.

(d) Give **one** reason why a reaction may not produce a single pure product.

**Answer:** 1 mark for any one: some reactant is left over / not all the reactants react; more than one product forms (a by-product / other product is mixed in); the reactants were impure; side reactions happen. Do not accept 'the scientist made a mistake' alone.
**Notes / working:** For example: "**Some of the reactants do not react**, so they are left mixed with the product." Other good answers: another product forms at the same time; the reactants were not pure.

**11** On a car journey, Mona's father explains why different metals are used in the car. The car body is made of steel (mostly iron) coated with a thin layer of zinc. **[1]**

(a) Which list puts the metals in order of reactivity, **most reactive first**?

- ☐ A) magnesium, zinc, iron, copper
- ☐ B) zinc, magnesium, copper, iron
- ☐ C) copper, iron, zinc, magnesium
- ☐ D) iron, zinc, magnesium, copper

**Answer:** A) magnesium, zinc, iron, copper
**Notes / working:** The reactivity series (most reactive first) includes: potassium, sodium, calcium, **magnesium, zinc, iron, copper**, silver, gold.

(b) Complete the word equation for the reaction of zinc with oxygen. **[1]**
zinc + oxygen → _____

**Answer:** zinc oxide or zincoxide or ZnO
**Notes / working:** zinc + oxygen → **zinc oxide**. A metal reacting with oxygen makes the metal oxide.

**12** (continued) **[1]**
Some electrical connectors in the car are coated with gold. Gold does not react with air, water or dilute acids, even after many years.

(c) Give the word that describes a substance that is generally unreactive.

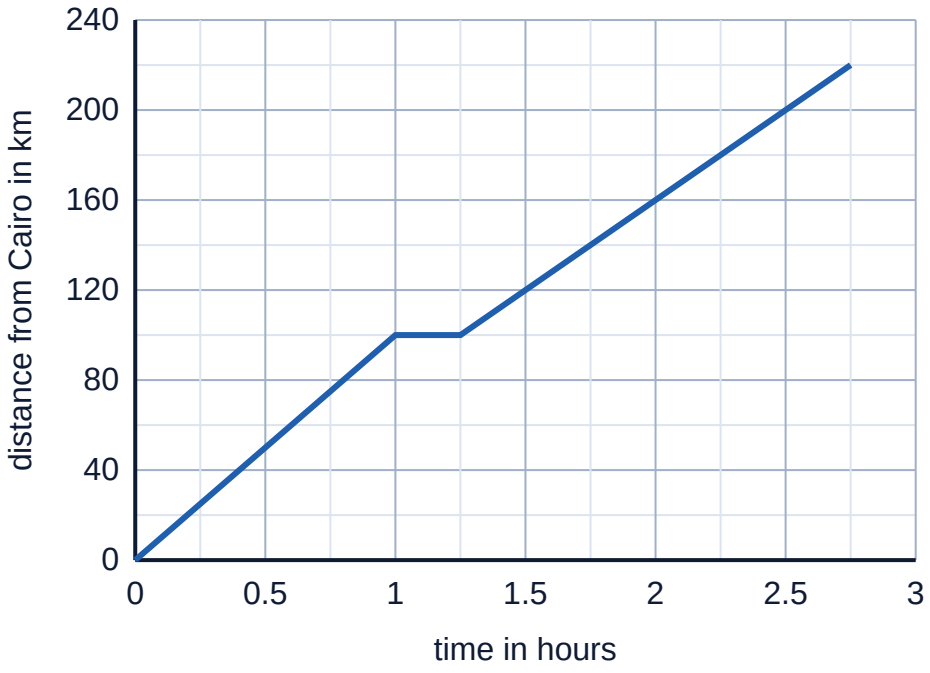
**Answer:** inert or an inert or noble
**Notes / working:** A substance that is generally unreactive is described as **inert**. Gold is used for connectors because it never corrodes, so the electrical contact stays good.

**13** (continued) **[1]**
Sodium is far too reactive to use in a car.

(d) Describe **one** thing you would see if a small piece of sodium were put into a trough of water.

**Answer:** 1 mark for any one: it floats; it fizzes / bubbles of gas form; it moves (whizzes) around on the surface; it melts into a ball; it gets smaller / disappears. Do not accept 'it explodes' or 'it sinks'.
**Notes / working:** Sodium **floats, fizzes and whizzes around** on the surface, melts into a shiny ball and gets smaller until it disappears. (It reacts to make sodium hydroxide and hydrogen gas.)

**14** Mona's family drives from Cairo along the Cairo–Alexandria Desert Road. They stop once at a rest house. The distance–time graph shows their journey. **[1]**




(a) Calculate the speed of the car during the **first hour**.

**Answer:** 100 km/h
**Notes / working:** In the first hour the car travels from 0 km to 100 km. speed = distance ÷ time = 100 ÷ 1 = **100 km/h**.

(b) The family reaches Alexandria after 2.75 hours. Calculate the **average speed** for the whole journey, including the stop. Show your working. **[2]**

**Answer:** 80 km/h
**Notes / working:** Total distance = 220 km (read from the end of the line). Total time = 2.75 h. Average speed = 220 ÷ 2.75 = **80 km/h**. Mark scheme: M1 for 220 ÷ 2.75 seen; A1 for 80. Common mistake: leaving out the stop and using 2.5 h (gives 88 km/h).

(c) On a straight part of the road, the car moves at a **constant speed**. Which statement about the forces on the car is correct? **[1]**

- ☐ A) The driving force is bigger than the friction and air resistance.
- ☐ B) The driving force is smaller than the friction and air resistance.
- ☐ C) The forces are balanced: the driving force equals the friction and air resistance.
- ☐ D) There are no forces acting on the car.

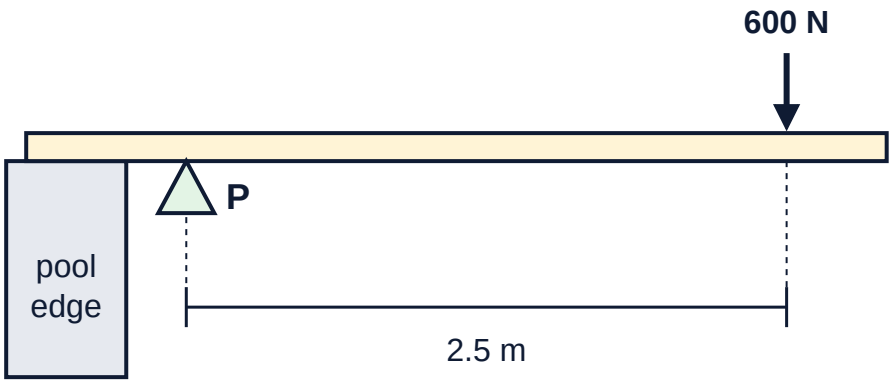
**Answer:** C) The forces are balanced: the driving force equals the friction and air resistance.
**Notes / working:** Constant speed in a straight line means the forces are **balanced**: forward force = backward forces. If the forward force were bigger, the car would speed up.

(d) Near Alexandria the driver presses the brakes and the car slows down. Explain why, in terms of forces. **[1]**

> **Answer:** 1 mark: the forces are unbalanced, with the backward force (friction/braking force) bigger than the forward (driving) force. Do not accept 'the brakes stop the car' with no forces.
> **Notes / working:** Model answer: "The forces become **unbalanced**: the **backward force** from the brakes and air resistance is **bigger than the forward force**, so the car slows down."

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**15** Youssef stands on the end of a diving board at the swimming pool. His weight is 600 N. The diagram shows a model of the board. The board can turn about point P. **[2]**




(a) Calculate the moment of Youssef's weight about P. Show your working.

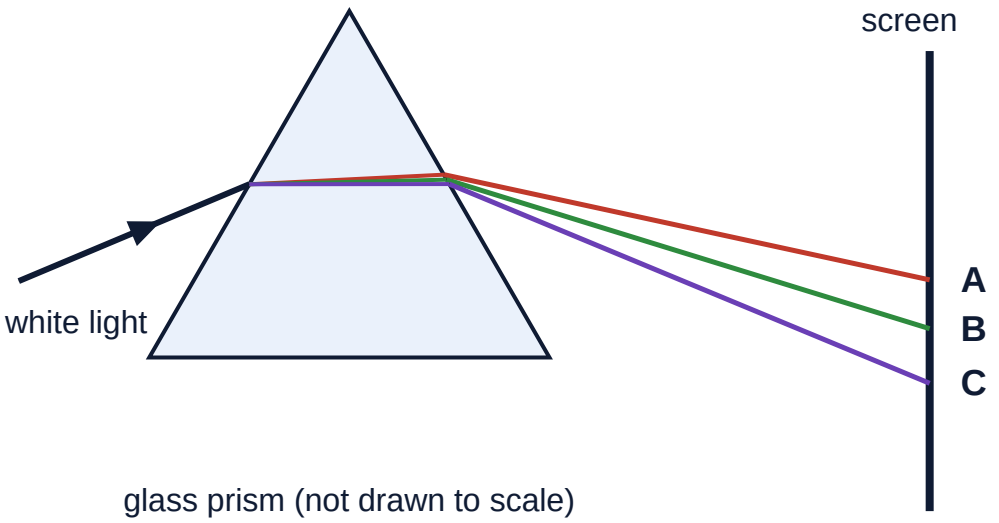
> **Answer:** 1500 N m
> **Notes / working:** moment = force × perpendicular distance from the pivot = 600 × 2.5 = **1500 N m**.
> Mark scheme: M1 for 600 × 2.5 seen; A1 for 1500 (N m).

(b) Youssef walks along the board **towards P**. What happens to the moment of his weight about P? **[1]**

- ☐ A) It increases.
- ☐ B) It stays the same.
- ☐ C) It changes direction.
- ☐ D) It decreases.

> **Answer:** D) It decreases.
> **Notes / working:** His weight stays 600 N, but the **distance from P gets smaller**, so moment = force × distance **decreases**.

**16** Amira is a photographer. She shines a narrow beam of white light through a glass prism onto a screen and photographs the colours. **[1]**




(a) Which letter, A, B or C, shows where **violet** light reaches the screen?

**Answer:** C
**Notes / working:** Violet light is refracted (bent) the **most** by the prism, so it ends up furthest from the original direction: **C**. Red is refracted least (A).

(b) Explain why light changes direction when it enters the glass prism. **[1]**

**Answer:** 1 mark: light slows down / changes speed when it passes from air into glass. Do not accept 'light bends' or 'the glass reflects it' on their own.
**Notes / working:** Light travels **more slowly in glass than in air**. When it crosses the boundary at an angle, the **change in speed** makes it change direction (refraction).

**17** (continued) **[1]**
Amira photographs a red apple with a green leaf on it. She puts a red filter in front of her camera lens.

(c) How do the apple and the leaf look in her photograph?

- ☐ A) The apple looks red and the leaf looks black.
- ☐ B) The apple looks red and the leaf looks green.
- ☐ C) The apple looks black and the leaf looks green.
- ☐ D) The apple and the leaf both look red.

**Answer:** A) The apple looks red and the leaf looks black.
**Notes / working:** The red filter lets only **red light** through. The red apple reflects red light, so it looks **red**. The green leaf reflects green light and absorbs red, so no light from the leaf passes through the filter: it looks **black**.

**18** In the food factory's laboratory, Karim investigates how the temperature of water affects diffusion. He adds one drop of food colouring to a beaker of still water and times how long the colour takes to spread evenly through the water. **[1]**

| Temperature of water in °C | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|
| Time to spread evenly in s | 330 | 250 | 290 | 130 | 90 |

(a) Which hypothesis can Karim test with this investigation?

- ☐ A) Food colouring tastes better in warm water.
- ☐ B) Particles of food colouring spread out faster in warmer water.
- ☐ C) Food colouring is made of particles that are too small to see.
- ☐ D) Diffusion is the most important process in the factory.

**Answer:** B) Particles of food colouring spread out faster in warmer water.
**Notes / working:** A **testable** hypothesis can be checked by measuring something. Karim measures time at different temperatures, so he can test whether the colour **spreads faster in warmer water**. Taste and 'most important' cannot be measured in this experiment, and the size of particles is not what he measures.

(b) What is the **dependent** variable? **[1]**

- ☐ A) the temperature of the water
- ☐ B) the volume of water in the beaker
- ☐ C) the size of the drop of food colouring
- ☐ D) the time for the colour to spread evenly

**Answer:** D) the time for the colour to spread evenly
**Notes / working:** The **dependent variable** is what Karim **measures**: the time. Temperature is the independent variable; the volume of water and size of drop are control variables.

(c) At which temperature is the result anomalous? **[1]**

**Answer:** 40 °C
**Notes / working:** The times should get shorter as the temperature rises. At **40 °C** the time (290 s) is longer than at 30 °C (250 s), so it does not fit the pattern.

(d) Describe the trend in the results and explain it using ideas about particles. **[2]**

**Answer:** 1 mark: as the temperature increases, the time decreases / the colour spreads (diffuses) faster. 1 mark: at higher temperature the particles (of water and colouring) move faster, so they intermingle / mix more quickly. Do not accept 'the particles get bigger' or 'the heat pushes the colour'.
**Notes / working:** Model answer: "**As the temperature increases, the time taken gets shorter**, so diffusion is faster. In warmer water the **particles move faster**, so the food colouring particles and the water particles **intermingle more quickly**."

(e) The water at 60 °C is a hazard. Describe the risk and **one** way to reduce it. **[1]**

> **Answer:** 1 mark for both: risk of burns / scalds (from hot water or a hot beaker) AND a sensible control, e.g. use heat-proof gloves or tongs, stand the beaker on a heatproof mat, do not move the beaker when full, wear eye protection, let it cool before emptying. Do not accept 'be careful' alone.
> **Notes / working:** Model answer: "Hot water could **scald** Karim's skin. He should **hold the beaker with heat-proof gloves or tongs** and keep it on a heatproof mat."

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**19** A large solar farm near Aswan uses thousands of solar panels to generate electricity. **[1]**

(a) Which list contains **only** renewable energy resources?

- ☐ A) solar, coal, wind
- ☐ B) solar, wind, tidal
- ☐ C) natural gas, oil, coal
- ☐ D) tidal, oil, solar

> **Answer:** B) solar, wind, tidal
> **Notes / working: Renewable** resources will not run out: solar, wind and tidal. Coal, oil and natural gas are fossil fuels, which are **non-renewable**.

(b) Explain **two** advantages of generating electricity from solar power near Aswan instead of by burning natural gas. **[2]**

> **Answer:** Any two, 1 mark each: Aswan has many hours of strong sunshine / few cloudy days; sunlight is renewable / will not run out (natural gas is non-renewable); no carbon dioxide (greenhouse gas) is released while generating, so it adds less to climate change; no air pollution (smoke) while generating; no fuel to buy. Do not accept 'it is cheaper' or 'it is better for the environment' with no reason.
> **Notes / working:** Model answer: "(1) Aswan gets **strong sunshine almost every day**, so the panels produce a lot of electricity. (2) Sunlight is **renewable**, and the panels give out **no carbon dioxide** while they work, but burning natural gas releases carbon dioxide and the gas will run out."

(c) Some of the packaging for the panels is made of **bioplastic**. What is a bioplastic? **[1]**

- ☐ A) a plastic made only from crude oil
- ☐ B) a metal that can bend like plastic
- ☐ C) a plastic that can never break down
- ☐ D) a plastic made from plant materials such as corn starch or sugar cane

> **Answer:** D) a plastic made from plant materials such as corn starch or sugar cane
> **Notes / working: Bioplastics** are made from plant materials (a renewable resource). Ordinary plastics are made from crude oil (non-renewable).

**20** Engineers at the solar farm study data about Aswan's weather and climate. **[1]**

(a) Which sentence describes Aswan's **climate**, not its weather?

- ☐ A) It was 41 °C in Aswan yesterday afternoon.
- ☐ B) There is a strong wind in Aswan this morning.
- ☐ C) Aswan is hot and very dry, with almost no rain in most years.
- ☐ D) It might rain in Aswan tomorrow.

**Answer:** C) Aswan is hot and very dry, with almost no rain in most years.
**Notes / working: Climate** is the typical (average) weather of a place over a long time (many years). **Weather** is the conditions at one time: yesterday, this morning, tomorrow.

(b) Scientists study tiny bubbles of air trapped in very old ice in Antarctica. What does this evidence show about the Earth's climate over the last 800 000 years? **[1]**

- ☐ A) It has changed many times between cold ice ages and warmer periods.
- ☐ B) It has stayed at the same temperature all the time.
- ☐ C) It has become warmer every single year.
- ☐ D) There was only one ice age, and it ended last century.

**Answer:** A) It has changed many times between cold ice ages and warmer periods.
**Notes / working:** Ice cores show that the Earth's climate moves in a **cycle between ice ages and warmer periods**, and each cycle takes a very long time (many thousands of years).

(c) Explain how burning fossil fuels can change the Earth's climate. **[2]**

**Answer:** 1 mark: burning fossil fuels releases carbon dioxide (a greenhouse gas) into the atmosphere. 1 mark: more heat (thermal energy) is trapped (by the atmosphere), so the Earth's average temperature rises / global warming. Do not accept 'it makes a hole in the ozone layer'.
**Notes / working:** Model answer: "Burning fossil fuels releases **carbon dioxide**, a **greenhouse gas**, into the atmosphere. More greenhouse gas **traps more heat**, so the Earth's average temperature rises and the climate changes."

**21** Aswan has very clear night skies. A school astronomy club looks at the Milky Way. **[1]**

(a) Which describes a **galaxy**?

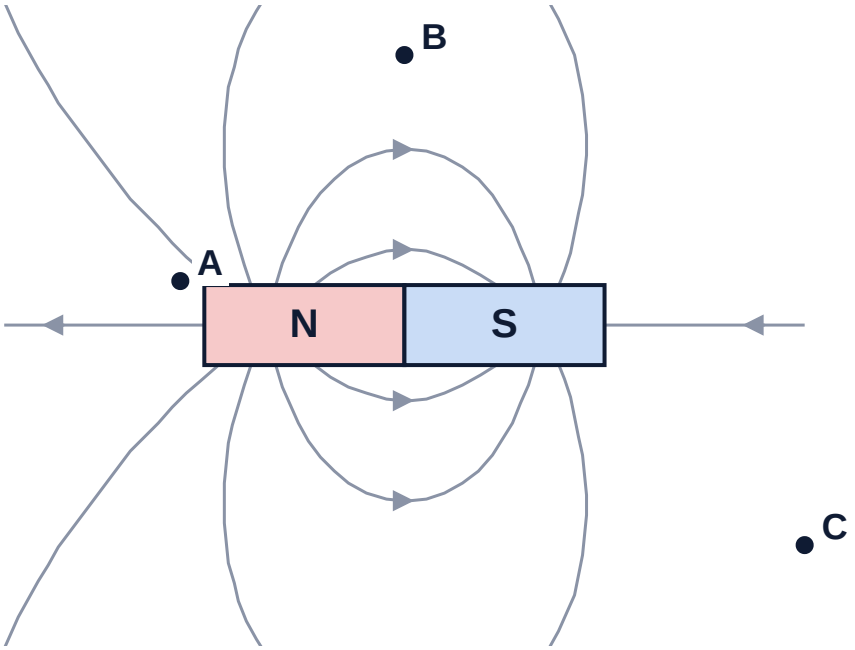
- ☐ A) one star with planets orbiting it
- ☐ B) a huge group of stars and planetary systems, with dust and gas, held together by gravity
- ☐ C) a cloud of gas around one planet
- ☐ D) a large rock that orbits the Sun

**Answer:** B) a huge group of stars and planetary systems, with dust and gas, held together by gravity
**Notes / working:** A **galaxy** contains billions of **stars**, many with **planetary systems**, plus **dust and gas**, all held together by gravity. One star with its planets is a planetary system (like our Solar System); a rock orbiting the Sun could be an asteroid.

(b) Describe how asteroids formed. **[1]**

**Answer:** 1 mark: from rocks / material left over when the planetary system (the planets / the Solar System) formed, which did not join together to make a planet. Do not accept 'broken pieces of stars' or 'from the Moon'.
**Notes / working:** Asteroids are rocks, smaller than planets. They are **rocks left over from the formation of the Solar System** that never joined together to make a planet.

**22** The diagram shows the magnetic field around a bar magnet. A, B and C are three points in the field. **[1]**




(a) At which point, A, B or C, is the magnetic field **strongest**?

**Answer:** A
**Notes / working:** The field is strongest where the field lines are **closest together**, near the poles. **A** is next to the N pole.

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**23** (continued) **[1]**
On the drive to Alexandria, Mona uses a compass to find north.

(b) Why does the Earth have a magnetic field?

- ☐ A) Sunlight magnetises the Earth's surface.
- ☐ B) The Moon pulls on the compass needle.
- ☐ C) The Earth's core acts like a magnet.
- ☐ D) The Earth's crust is made of bar magnets.

**Answer:** C) The Earth's core acts like a magnet.
**Notes / working:** The Earth has a magnetic field because its **core acts as a magnet**. The magnetic field exerts a force on the compass needle (a small magnet), so the needle lines up north–south.